

Wanglong Lu

Senior Data Scientist | Applied Multimodal Intelligence

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Research & Professional Profile

Applied multimodal intelligence researcher and Senior Data Scientist at Nasdaq, with a Ph.D. in Computer Science. Research spans high-resolution generative image editing, multimodal control, image and document restoration, efficient adaptation, and applied machine learning. Industry experience covers financial ML pipelines, feature selection, internal ML platform deployment, cloud-based experimentation, and AI-assisted engineering tools.

Professional Experience

Nasdaq / Nasdaq Verafin

St. John's, NL, Canada

Senior Data Scientist, AI/Analytics

Mar. 2025 – Present

- Contribute application deployment and integration to a **cross-team ML platform under development**, whose scope spans data ingestion, preprocessing, distributed training, feature selection, evaluation, and an auditable model registry.
- Deployed the **existing UI/backend prototype** into an internal workspace, configuring network connectivity and IAM roles for internal access.
- Implemented **PostgreSQL-backed persistence for user state** and deployed **FastAPI services on AWS Lambda using Mangum**; integrated these components with the platform's application layer.
- Designed cross-scale ensemble feature selection that removed **77.2% of input features** while retaining competitive performance on CFML data; presented the algorithm to more than 60 attendees.
- Built **SageMaker Job Submitter**, a Python package for AWS SageMaker Processing, Training, and Hyperparameter Tuning jobs, including automated dependency packaging and SageMaker Pipelines. Package-based submission cut developer job-configuration time from **7 days to 1 day**, an approximately **86% reduction**.
- Developed an AWS Bedrock-based feature-formula validation engine with structured logical-consistency checks and reliability scores to flag hard-coded errors.
- Built a **PagerDuty RAG incident-response assistant** as an internal hackathon project on the company's GenAI platform; used incident context to retrieve relevant troubleshooting guidance and resolution steps for on-call engineers.
- Mentored industry research on cheque OCR quality assessment and text extraction, and tabular-data generation, supporting pipeline design and generative-model architecture.

AI Algorithm Intern

May 2024 – Jan. 2025

- Designed diffusion-based text-document restoration and inpainting for degraded cheque images using Python and PyTorch; benchmarked against existing methods with AWS, SLURM, CUDA, and shell scripts.
- Designed an image-verification algorithm using patch-level Gaussian latent modeling; presented research progress to 39 cross-functional attendees at the end of the internship.

AI Intern – Beaufort Solutions

Jul. 2022 – Nov. 2022

St. John's, NL, Canada

- Designed a data-collection pipeline and few-shot, vision-language image-theme recognition for personalized photo-book curation; contributed to TEG (Journal of Electronic Imaging, 2024).

AI Algorithm Intern – Zhongkong Hanlian

Dec. 2017 – Aug. 2018

Hangzhou, China

- Built vehicle-logo recognition using deep networks and developed annotation tools in Python, C++, and Java; supported real-time deployment at community security gates.

Education

Ph.D., Computer Science – Memorial University of Newfoundland, Canada	2021–2025
M.Sc., Computer Software and Theory – Wenzhou University, China	2018–2021
B.Sc., Digital Media Technology – Communication University of Zhejiang, China	2014–2018

Selected Research & Engineering Projects

UltraDiffEdit

TNNLS 2026; first author

Tuning-free high-resolution generative image editing

- Developed multi-patch encoding, global-local consistency denoising, and progressive sampling to extend pretrained latent diffusion models to real-image editing without fine-tuning.
- Demonstrated editing up to 8K resolution on a single RTX 3090. Code and video are available.
- Built three editing benchmarks (DIV2KEdit, Syn2KEdit, UHRSEdit) for high-resolution, multimodal evaluation; tested pretrained SDXL and ControlNet models without fine-tuning.

FACEMUG

TVCG 2024; first author

Multimodal local facial editing

- Developed a unified editing framework for text, sketches, semantic maps, color maps, exemplars, and attribute labels, with multimodal feature fusion and self-supervised latent warping.
- Preserves the unedited image regions while supporting local, controllable changes. Presented the work through the Genesis entrepreneur training program. Publication.

AdaMSS

NeurIPS 2025; co-author

Parameter-efficient fine-tuning

- Co-authored adaptive multi-subspace updates for expressive and efficient adaptation of pretrained language and vision models.
- The method is integrated into Hugging Face PEFT. Research code is public.

DocPure

TCSVT 2026; co-first author

Prompt-free unified document restoration

- Co-authored structure-guided wavelet modulation for restoration across document degradations without manual inference-time task prompts. Code.
- The unified model was evaluated on 12 datasets across deblurring, denoising, compression artifact reduction, and deshadowing. Benchmarks.

GRIG

CVM 2025; first author

Data-efficient generative residual image inpainting

- Developed iterative residual inpainting using CNN features and Transformer reasoning for small training sets.
- Introduced forgery-patch adversarial training to improve textures and fine detail. Paper; code.

Visual Style Prompt Learning

Pattern Recognition 2025; first author

Diffusion-guided blind face restoration

- Developed diffusion-generated visual style prompts to guide blind face restoration.
- Designed style-modulated aggregation to fuse visual prompts with restoration features. Paper; code.

Local Agentic Code Editor

Nasdaq internal prototype; 2026

Exploratory side project: API-driven code editing and local tool orchestration

- Prototyped a local AI code editor with an **agentic loop** that selects between code writing, code review, and shell-tool execution based on the current task.
- Integrated **language-model APIs with local tool execution** to support code generation, debugging, and iterative review.

Peer-Reviewed Publications

Tuning-Free Latent Diffusion Models for Ultrahigh-Resolution Image Editing

Wanglong Lu, Lingming Su, Kaijie Shi, Minglun Gong, Xiaogang Jin, Hanli Zhao, Xianta Jiang

IEEE Transactions on Neural Networks and Learning Systems, 1-15, 2026

UHDRes: Ultra-High-Definition Image Restoration via Dual-Domain Decoupled Spectral Modulation

Shihao Zhao, Wanglong Lu, Binhao Wang, Tao Wang, Kaihao Zhang, Hanli Zhao

IEEE Transactions on Circuits and Systems for Video Technology, 2026

DocPure: Prompt-Free Unified Document Restoration via Degradation-Aware Structure-Guided Wavelet Modulation

Lingming Su, Wanglong Lu, Tao Wang, Kaihao Zhang, Nan Zhang, Liyan An, Hanli Zhao

IEEE Transactions on Circuits and Systems for Video Technology, 2026

EchoSR: Efficient Context Harnessing for Lightweight Image Super-Resolution

Hanli Zhao, Binhao Wang, Shihao Zhao, Tao Wang, Kaihao Zhang, Wanglong Lu

Information Fusion, 104471, 2026

AdaMSS: Adaptive Multi-Subspace Approach for Parameter-Efficient Fine-Tuning

Jingjing Zheng, Wanglong Lu, Yiming Dong, Chaojie Ji, Yankai Cao, Zhouchen Lin

NeurIPS 2025

Visual style prompt learning using diffusion models for blind face restoration

Wanglong Lu, Jikai Wang, Tao Wang, Kaihao Zhang, Xianta Jiang, Hanli Zhao

Pattern Recognition, 2025

Toward Biosignals-Free Autonomous Prosthetic Hand Control via Imitation Learning

Kaijie Shi, Wanglong Lu, Hanli Zhao, Vinicius Prado da Fonseca, Ting Zou, Xianta Jiang

IEEE Transactions on Neural Systems and Rehabilitation Engineering 33, 3544-3554, 2025

GRIG: Data-Efficient Generative Residual Image Inpainting

Wanglong Lu, Xianta Jiang, Xiaogang Jin, Yong-Liang Yang, Minglun Gong, Kaijie Shi, Tao Wang, Hanli Zhao

Computational Visual Media 11(6), 1329-1361, 2025

Do Inpainting Yourself: Generative Facial Inpainting Guided by Exemplars

Wanglong Lu, Hanli Zhao, Xianta Jiang, Xiaogang Jin, Yongliang Yang, Min Wang, Jiankai Lyu, and Kaijie Shi

Neurocomputing 617, 128996, 2025

Degradation-Aware Frequency-Separated Transformer for Blind Super-Resolution

Hanli Zhao, Binhao Wang, Wanglong Lu, Juncong Lin

International Conference on Computational Visual Media, 2025

Real-time dual-eye collaborative eyeblink detection with contrastive learning

Hanli Zhao, Yu Wang, Wanglong Lu, Zili Yi, Jun Liu, Minglun Gong

Pattern Recognition, 2025

FACEMUG: A Multimodal Generative and Fusion Framework for Local Facial Editing

Wanglong Lu, Jikai Wang, Xiaogang Jin, Xianta Jiang, and Hanli Zhao

IEEE Transactions on Visualization and Computer Graphics, 2024

Handling the Non-smooth Challenge in Tensor SVD: A Multi-objective Tensor Recovery Framework

Jingjing Zheng, Wanglong Lu, Wenzhe Wang, Yankai Cao, Xiaoqin Zhang, and Xianta Jiang

ECCV, 2024

Restoring Vision in Hazy Weather with Hierarchical Contrastive Learning

Tao Wang, Guangpin Tao, Wanglong Lu, Kaihao Zhang, Wenhan Luo, Xiaoqin Zhang, Tong Lu.

Pattern Recognition Volume 145, January 2024, 109956

TEG: image theme recognition using text-embedding-guided few-shot adaptation

Jikai Wang, Wanglong Lu, Yu Wang, Kaijie Shi, Xianta Jiang, Hanli Zhao

Journal of Electronic Imaging 33(1), 013028, 2024

Detecting Blinks from Wearable Cameras using Spatial-Temporal-Aware Deep Network Learning

Yu Wang, **Wanglong Lu**, Hanli Zhao, Xianta Jiang, Bin Zheng, M. Stella Atkins.

ETRA 23, 2023

Efficient Generative Face Completion with Perceptual Deblurring

Hanli Zhao, Ying Liu, **Wanglong Lu**, Xiaogang Jin, Hui Huang, Kaijie Shi

Journal of Computer-Aided Design & Computer Graphics 34(9), 1420-1431, 2022

Generative image inpainting with enhanced gated convolution and Transformers

Min Wang, **Wanglong Lu**, Jiankai Lyu, Kaijie Shi, Hanli Zhao.

Displays 76, 102321, 2022

Probability-Based Channel Pruning for Depthwise Separable Convolutional Networks

Han-Li Zhao, Kai-Jie Shi, Xiaogang Jin, Ming-Liang Xu, Hui Huang, **Wanglong Lu**, Ying Liu.

Journal of Computer Science and Technology 37, 584-600, 2022

Image Splicing Tamper Detection Based on Multi-Scale Feature Priors [in Chinese; translated title]

Wanglong Lu et al.

China Sciencepaper 17(11), 2022

Category-consistent deep network learning for accurate vehicle logo recognition

Wanglong Lu, Hanli Zhao, Qi He, Hui Huang, Xiaogang Jin.

Neurocomputing 463, 623-636, 2021

High-quality retinal vessel segmentation using generative adversarial network with a large receptive field

Hanli Zhao, Xiaqing Qiu, **Wanglong Lu**, Hui Huang, Xiaogang Jin.

International Journal of Imaging Systems and Technology 30(3), 828-842, 2020

Preprints

Simulation-Driven Imitation Learning for Biosignals-Free Shared-Autonomy Prosthetic Grasping

Kaijie Shi, **Wanglong Lu**, Han Chen, Vinicius Prado da Fonseca, Ting Zou, Hanli Zhao, Xianta Jiang

arXiv preprint arXiv:2606.07389, 2026

TextDoctor: Unified Document Image Inpainting via Patch Pyramid Diffusion Models

Wanglong Lu, Lingming Su, Jingjing Zheng, Vinícius Veloso de Melo, Farzaneh Shoeleh, John Hawkin, Terrence Tricco,

Hanli Zhao, Xianta Jiang

ArXiv, 2025

StDDPM: Sequential Financial Synthetic Data Generation using T-distribution based Diffusion Model

Omid Tarkhaneh, Hamideh Mehri, **Wanglong Lu**, Farzaneh Shoeleh, Vinicius Veloso de Melo

Research Square preprint, 2025

Promptrr: Diffusion models as prompt generators for single image reflection removal

Tao Wang, **Wanglong Lu**, Kaihao Zhang, Wenhan Luo, Tae-Kyun Kim, Tong Lu, Hongdong Li, Ming-Hsuan Yang

ArXiv, 2024

Dissertation

Generative Models for Semantic Facial Image Editing: Multimodal Approaches

Wanglong Lu

PhD thesis, Memorial University of Newfoundland, 2025

Technical Skills

Programming: Python, PyTorch, SQL, Git, Bash, C++, Java, CUDA, OpenCV.

ML & GenAI: Latent diffusion, GANs, Transformers, multimodal conditioning, vision-language models, PEFT/LoRA, Hugging Face, RAG, feature selection, experimental evaluation.

Infrastructure: FastAPI, Mangum, PostgreSQL; AWS SageMaker, Bedrock, Lambda, S3, Docker, GPU experimentation, SLURM.

Additional Engineering Project

AI Headshot Generation Deployment: Implemented model invocation and application deployment for AI-generated headshots.

Teaching, Mentorship & Talks

- Mentored 10 students on research projects spanning generative vision, restoration, image recognition, and applied machine learning.
- Teaching assistant, Memorial University: Data Structures and Algorithms (Sep.–Dec. 2021); Introduction to Machine Learning (Jan.–Apr. 2024).
- Guest lecturer, Bio-Inspired Robotics (ENGI 9986), Memorial University, Feb. 2025: a 150-minute lecture on Deep Generative Models and Applications for 10 graduate students.
- Invited speaker, IEEE Newfoundland and Labrador Section AGM, Mar. 2026: Generative Models for Semantic Image Editing: Multimodal Approaches.
- Research and technical presentations to academic and industry audiences, including the 2026 feature-selection presentation to 60+ attendees.

Academic Service

Invited reviewer, 2026: IEEE TPAMI, IEEE TMM, Pattern Recognition, Applied Soft Computing, Engineering Applications of Artificial Intelligence, Scientific Reports, Neurocomputing, Knowledge-Based Systems, and ECCV 2026.

2025: IEEE TIP, IEEE TMM, Pattern Recognition, Neurocomputing, CAD/Graphics, Advances in Manufacturing, Signal, Image and Video Processing, IEEE Signal Processing Letters, Multimedia Systems, Journal of Machine Learning and Cybernetics.

2024: IEEE TMM, IEEE TCSVT, Knowledge-Based Systems, Journal of Medical Systems, Cluster Computing, Journal of Visual Communication and Image Representation, Displays, The Visual Computer, and ETRA PETMEI.

Granted Patents

Co-inventor on four Chinese invention patents:

- Category-consistent deep network learning for image recognition. ZL 2021 1 0408724.X.
- Deep feature-based convolutional neural network for vehicle-logo recognition. ZL 2020 1 0139043.3.
- GAN-based intelligent segmentation of retinal blood-vessel images. ZL 2019 1 0884346.5.
- Convolutional neural network-based intelligent vehicle-logo detection. ZL 2020 1 0139068.3.

Selected Honors & Awards

- Computer Innovation and Entrepreneurship Award, Wenzhou Computer Federation; first prize, team leader (2024).
- Genesis entrepreneur training program, Memorial University (2024).
- Outstanding Research Award (Ph.D.), Computer Science, Memorial University (2023).
- Mitacs Accelerate Award (2022).
- Outstanding Graduate, Wenzhou University (2020); Outstanding Graduate of Zhejiang Province (2018).
- Chinese Graduate Mathematical Modeling Competition, third prize (2019); Lanqiao Cup C/C++ Programming Competition, second prize (2019); CCPC regional competition, third prize (2019).